

Private AWS EC2 Behind Cloudflare Tunnel

Project Screenshots — Syed Basit Aftab | github.com/saftab4-arch

Architecture: Private EC2 instance (no public IP, no key pair) in a custom VPC, accessed via AWS SSM Session Manager. Nginx serves a custom webpage. Cloudflare Tunnel (cloudflared) exposes the site publicly at **web.basitcloudlab.com** — no inbound security group rules required.

01. VPC + Subnet + Internet Gateway + Route Table
02. IAM Role (EC2-SSM-Role) for SSM access
03. EC2 instance (no public IP, no key pair)
04. 3× VPC Interface Endpoints (ssm, ssmmessages, ec2messages)
05. SSM Session Manager connection verified
06. Elastic IP allocated (unused — tunnel handles ingress)
07. Nginx installed + custom webpage deployed
08. Cloudflare Zero Trust tunnel created (basit-cloud-tunnel)
09. cloudflared installed and running as systemd service
10. web.basitcloudlab.com publicly accessible via Cloudflare

01 - VPC Created (basit-cloud-vpc, 10.0.0.0/16)

You successfully created vpc-02f9f83b3efed8a86 / basit-cloud-vpc

Your VPCs

VPCsVPC encryption controls

Your VPCs (2) info

Find VPCs by attribute or tag

Last updated less than a minute ago

Actions

Create V

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP option set	Main route table	Main network ACL
☐	-	vpc-057247e48a6a699de	Available	-	-	Off	172.31.0.0/16	-	dopt-0f834e1c71632bec4	rtb-090d94d2197cf2c52	acl-05086e47c272714bc
☐	basit-cloud-vpc	vpc-02f9f83b3efed8a86	Available	-	-	Off	10.0.0.0/16	-	dopt-0f834e1c71632bec4	-	-

02 - Public Subnet Created (public-subnet-1, 10.0.1.0/24)

You have successfully created 1 subnet: subnet-06b5117dd0fcd35

Subnets (1/1) Info

Last updated less than a minute ago

Actions

Create subnet

Find subnets by attribute or tag

Subnet ID : subnet-06b5117dd0fcd35

Clear filters

☒	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6 CIDR association ID	Available IPv4 addresses	Availability Zone
☒	public-subnet-1	subnet-06b5117dd0fcd35	Available	vpc-02f9f83b3efed8a86 basit...	Off	10.0.1.0/24	--	--	251	us-east-1a

03 - Internet Gateway Created and Attached (basit-cloud-igw)

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#InternetGatewayinternetGatewayId=igw-0a0c94d5bc12e413f

Search [Alt+S]

United States (N. Virginia) basit-devops126 (587769523903) igw-admin

VPC > Internet gateways > igw-0a0c94d5bc12e413f

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways**
- Egress-only Internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IP
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

Security

- Network ACLs
- Security groups

PrivateLink and Lattice

- Getting started
- Endpoints

Internet gateway igw-0a0c94d5bc12e413f successfully attached to vpc-02f9f83b3efed8a86

igw-0a0c94d5bc12e413f / basit-cloud-igw [Actions](#)

Details [Info](#)

Internet gateway ID igw-0a0c94d5bc12e413f	State Attached	VPC ID vpc-02f9f83b3efed8a86 / basit-cloud-vpc	Owner 587769523903
--	--------------------------------	---	---------------------------------------

Tags (1) [Manage tags](#)

Search tags

Key	Value
Name	basit-cloud-igw

04 - Route Table Updated and Associated (basit-cloud-rt)

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#RouteTableDetails:RouteTableId=rtb-0580963c7d53ea026

VPC > Route tables > rtb-0580963c7d53ea026

Updated routes for rtb-0580963c7d53ea026 / basit-cloud-rt successfully

Details

rtb-0580963c7d53ea026 / basit-cloud-rt

Actions

Details
Route table ID
rtb-0580963c7d53ea026
VPC
vpc-02f9f83b5efed8a86 | basit-cloud-vpc

Main
No
Owner ID
587769523903

Explicit subnet associations
subnet-06b5117dddfdc35 / public-subnet-1

Edge associations
-

Routes Subnet associations Edge associations Route propagation Tags

Routes (2)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0a0c94d5bc12e413f	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

05 - IAM Role Created (EC2-SSM-Role)

AM > Roles

Identity and Access Management (IAM)

Search IAM

Dashboard

Groups Management

- Groups
- Users
- User groups
- Identity providers
- Account settings
- Access management
- Temporary delegation requests

Reports

- AWS Security Analyzer
- Source analysis
- Trusted access
- Analyzer settings
- Initial report
- Organization activity
- Access control policies
- Resource control policies

Related consoles

Identity Center

Organizations

Role EC2-SSM-Role created.

Roles (11)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Search

☐	Role name	Trusted entities	Last activity
☐	AWSServiceRoleForAutoScaling	AWS Service: autoscaling (Service-Linked Role)	28 days ago
☐	AWSServiceRoleForElasticLoadBalancing	AWS Service: elasticloadbalancing (Service-Linked Role)	24 days ago
☐	AWSServiceRoleForOrganizations	AWS Service: organizations (Service-Linked Role)	-
☐	AWSServiceRoleForRDS	AWS Service: rds (Service-Linked Role)	18 minutes ago
☐	AWSServiceRoleForResourceExplorer	AWS Service: resource-explorer-2 (Service-Linked Role)	16 minutes ago
☐	AWSServiceRoleForSSO	AWS Service: sso (Service-Linked Role)	8 hours ago
☐	AWSServiceRoleForSupport	AWS Service: support (Service-Linked Role)	-
☐	AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service-Linked Role)	-
☐	AWSServiceRoleForVPCTransitGateway	AWS Service: transitgateway (Service-Linked Role)	9 days ago
☐	EC2-SSM-Role	AWS Service: ec2	-
☐	test-function-role-3p1f0m6f	AWS Service: lambda	42 days ago

Roles Anywhere

Authenticate your non AWS workloads and securely provide access to AWS services.



Access AWS from your non AWS workloads

Operate your non AWS workloads using the same authentication and authorization strategy that you use within AWS.



X.509 Standard

Use your own existing PKI infrastructure or use [AWS Certificate Manager Private Certificate Authority](#) to authenticate identities.

06 - EC2 Instance Launched (cloudflare-private-web, t2.micro)

EC2 > Instances

Instances (1) info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Save filter sets

Choose filter set

Find instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Security group name	Key name	Launch time
☐	cloudflare-priv...	i-0afafadcb1d64d2e5	Running	t2.micro	-	us-east-1a	-	-	-	-	disabled	private-web-sg	-	2025/05/30 20:57

07 - VPC Endpoint Created: com.amazonaws.us-east-1.ssm

Successfully created VPC endpoint
vpce-0167759d01e8e0b65

Endpoints (1/1) info

Find endpoints by attribute or tag

vpce-0167759d01e8e0b65 Clear filters

	Name	VPC endpoint ID	Endpoint type	Status	Service name	Service network	Resource configuration	Service region
☒	my-endpointCF	vpce-0167759d01e8e0b65	Interface	Pending	com.amazonaws.us-east-1.ssm	-	-	us-east-1

08 - VPC Endpoint Created: com.amazonaws.us-east-1.ssmmessages

Successfully created VPC endpoint
vpce-0d92b7a5f0fb1dd83

Endpoints (2) info

Find endpoints by attribute or tag

☐	Name	VPC endpoint ID	Endpoint type	Status	Service name	Service network	Resource configuration	Service region
☐	my-endpointCF1	vpce-0d92b7a5f0fb1dd83	Interface	Pending	com.amazonaws.us-east-1.ssmmessages	-	-	us-east-1
☐	my-endpointCF	vpce-0167759d01e8e0b65	Interface	Available	com.amazonaws.us-east-1.ssm	-	-	us-east-1

09 - VPC Endpoint Created: com.amazonaws.us-east-1.ec2messages

Successfully created VPC endpoint
vpce-08aca9e83e69d2ba2

Endpoints (1/1) Info

Find endpoints by attribute or tag

VPCE endpoint ID: vpce-08aca9e83e69d2ba2 Clear filters

	Name	VPCE endpoint ID	Endpoint type	Status	Service name	Service network	Resource configuration	Service region
☒	my-endpointCF2	vpce-08aca9e83e69d2ba2	Interface	Pending	com.amazonaws.us-east-1.ec2messages	-	-	us-east-1

10 - SSM Session Manager Connected (private IP: 10.0.1.241)

1. **Identifikasi Masalah**: Menentukan masalah yang akan diteliti.

```

sh-5.2$ whoami
asm-user
sh-5.2$ ifconfig
enX0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 9001
    inet 10.0.1.241 netmask 255.255.255.0 broadcast 10.0.1.255
    inet6 fe80::9d:46ff:fefd:f6d5 prefixlen 64 scopeid 0x20<link>
    ether 02:9d:46:fd:f6:d5 txqueuelen 1000 (Ethernet)
    RX packets 1579 bytes 247342 (241.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2062 bytes 268579 (262.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 12 bytes 1020 (1020.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 12 bytes 1020 (1020.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

sh-5.2$

```

11 - Elastic IP Allocated (98.82.220.57)

SSRS

Associate this Elastic IP address

Allocate Elastic IP address

[Clear filters](#)

	Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Associated instance ID	Private IP address	Association ID	Network interface owner account	Network border
☐	-	98.82.220.57	Public IP	eipalloc-005e4874b1b31e518	-	-	-	-	-	us-east-1

	-	98.82.220.57	Public IP	eipalloc-005e4874b1b31e518	-	-	-	-	-	us-east-1
---	---	------------------------------	-----------	----------------------------	---	---	---	---	---	-----------

12 - Nginx Running (Welcome to nginx! via 100.50.234.82)



Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

13 - Custom Web Page Served (Basit Cloud Lab via Nginx)

← → ↻ ⚠ Not secure 100.50.234.82

Basit Cloud Lab

Private AWS EC2 behind Cloudflare Tunnel

Built by Syed Basit Aftab

14 - Cloudflare Zero Trust Dashboard

[← Back to tunnels](#)

basit-cloud-tunnel

[Overview](#) [CIDR routes](#) [Hostname routes](#) [Published application routes](#) [Live logs](#)

[All](#) [Basic Information](#) [Connectors](#)

Basic Information

Tunnel name	Tunnel type	Tunnel ID
basit-cloud-tunnel Edit	cloudflared	52f63bc8-b164-47c5-8008-7cab1ef0a439 Copy
Status	Uptime	
HEALTHY	4 minutes	

Connectors

Showing 1 - 1

Review established connections between Cloudflare's network and your infrastructure.

[Add a connector](#)

Search a connector

Connector ID	Hostname	Data centers	Origin IP	Version	Platform	Status
3817617e-6810-4ceb-a85a-dfa4ff6dc25c2 Copy	ip-10-0-1-241ec2.internal	iad	100.50.234.82	2026.5.2	linux_amd64	Connected

1 - 1 Items per page: 10

< 1 of 1 page

15 - Cloudflare DNS Record: web.basitcloudlab.com → Tunnel

DNS records for basitcloudlab.com

Manage how the Internet finds your web content, verifies services, and routes traffic.

DNS Setup: Full

Share feedback

Recommendations

About DNS

Proxied vs. DNS only

TTL & caching

- Visitors cannot reach www.basitcloudlab.com.
Add an A, AAAA, or CNAME record for www and optionally create a [redirect rule](#) to send visitors to basitcloudlab.com.
- Visitors cannot reach basitcloudlab.com.

Show all 3 recommendations

Search DNS Records

Filters

Display options

Import

Export

+ Add record

1 records

☐	☐	☐	Name ☐	Type ☐	Content ☐	Proxy status ☐	TTL ☐	Tags	Comment	Details
☐			web.basitcloudlab.com	Tunnel	basit-cloud-tunnel	 Proxied	Auto	—	—	—

Showing 1-1 of 1

16 - Cloudflare Tunnel Error 1033 (tunnel down before fix)

The screenshot displays the Cloudflare One dashboard for the account 'Saftab4@wgu.edu'. The interface includes a left-hand navigation menu with options like Overview, Insights & Logs, Team & Resources, Networks, Access controls, and Settings. The main content area is titled 'Cloudflare One' and provides a 'Get started with Zero Trust' guide. It lists several recommended actions: replacing a client-based VPN, securing web traffic from advanced threats, setting up secure access to private apps, and securing the network with DNS filtering. Below this, the 'Manage your account' section shows details for the team name, billing plan (Zero Trust Free), and login page customization. The bottom of the image shows a Windows taskbar with a weather widget indicating 55°F and a search bar.

Saftab4@wgu.edu's ...

Quick search... ctrl + K

Back to Saftab4@wgu.edu'...

Overview

Insights & Logs

Team & Resources

Networks

Overview

Connectors

Routes

Resolvers & Proxies

Access controls

Traffic policies

Cloud & SaaS findings

Email security

Data loss prevention

Browser isolation

Reusable components

Integrations

Settings

Give feedback

Cloudflare One

Connect and secure your users, networks, agents, and data with Zero Trust security. See account health and actionable insights over the past seven days.

Get started with Zero Trust

Set up your account, find helpful resources, and explore what to do next.

All Recommendations Account settings Resources and support

What would you like to do next?

- Replace my client-based or site-to-site VPN**
Connect networks and devices without the security gaps of legacy tools.
- Secure my web traffic from advanced threats**
Enforce security policies to stop threats before they reach your network.
- Set up secure access to private apps from any browser**
Provide simple, per-resource access to private services that use HTTP, RDP, or SSH.
- Secure my network with DNS filtering**
Block threats at the source to increase security without adding latency.

Manage your account

- Team name: red-art-31ee**
Your team name is used to set up identity providers, find applications behind your App Launcher, and connect devices via the device client.
- Billing plan: Zero Trust Free**
Change your plan type, purchase additional seats or add-ons, or update your payment method.
- Customize your login page**
Customize the page displayed to end users when they reach an Access application or launch an isolated browser.

55°F Clear

Search

17 - cloudflared Service Active and Running (systemctl status)

```
sh-5.25 sudo systemctl stop cloudflared
sh-5.25 sudo systemctl start cloudflared
Job for cloudflared.service failed because a timeout was exceeded.
See "systemctl status cloudflared.service" and "journalctl -xeu cloudflared.service" for details.
sh-5.25 ip route
default via 10.0.0.1 dev enx0 proto dhcp src 10.0.0.1 metric 512
10.0.0.1 via 10.0.0.1 dev enx0 proto dhcp src 10.0.0.1 metric 512
10.0.0.2 dev enx0 proto kernel scope link src 10.0.0.2 metric 512
10.0.0.1 dev enx0 proto dhcp scope link src 10.0.0.1 metric 512
sh-5.25 curl -I https://cloudflare.com
HTTP/2 200
sh-5.25 sudo systemctl start cloudflared
sudo systemctl status cloudflared
curl -I https://cloudflare.com
# cloudflared.service - cloudflared
Loaded: loaded (/etc/systemd/system/cloudflared.service; enabled; preset: disabled)
Active: active (running) since Sun 2024-05-31 02:04:42 UTC; 56 s ago
Main PID: 2636 (cloudflared)
Tasks: 6 (limit: 1120)
Memory: 17.7M
CPU: 24ms
CGroup: /system.slice/cloudflared.service
└─636 /usr/bin/cloudflared --no-autoupdate tunnel run --token eyJhbGciOiJIUzI1NiIsInR5cGEiOiJ1bmEiLCJpdjkiOiJ1cm9udCI7fQoIbnNjbHViYyI6ImNlNm81dlwycyI6IkltRTFFaVppdWpjcDExSVF0cAwWIiwiaXNjaXNlaXNlaXR5b29tOTBmZSI6bnNldkIiwiaWF0eC99
May 31 02:04:42 ip-10-0-0-241.ec2.internal cloudflared[2636]: 2026-05-31T02:04:42 INF Tunnel connection curve preferences {K25519MLKM768 CurveID(45074) CurveP256} connIndex=0 event=0 ip=198.41.200.33
May 31 02:04:42 ip-10-0-0-241.ec2.internal cloudflared[2636]: 2026-05-31T02:04:42 INF Registered tunnel connection connIndex=0 connection=b49ccle-c027-4682-b002-65lec0b8ee9 event=0 ip=198.41.200.33 location=iad20 protocol=quic
May 31 02:04:42 ip-10-0-0-241.ec2.internal systemd[1]: Started cloudflared.service - cloudflared.
May 31 02:04:42 ip-10-0-0-241.ec2.internal cloudflared[2636]: 2026-05-31T02:04:42 INF Tunnel connection curve preferences {K25519MLKM768 CurveID(45074) CurveP256} connIndex=1 event=0 ip=198.41.192.57
May 31 02:04:42 ip-10-0-0-241.ec2.internal cloudflared[2636]: 2026-05-31T02:04:42 INF Updated to new configuration config="{\"ingress\":[{\"hostname\":\"web.basilcloudlab.com\",\"service\":{\"http://localhost:80/},{\"https://localho\"}]},\"routing\":{\"\"ip-10-0-0-241.ec2.internal cloudflared[2636]: 2026-05-31T02:04:42 INF Registered tunnel connection connIndex=1 connection=d097ba2-fals-fals-d43-fdb0c0bb0bb event=0 ip=198.41.192.57 location=iad16 protocol=quic
May 31 02:04:43 ip-10-0-0-241.ec2.internal cloudflared[2636]: 2026-05-31T02:04:43 INF Tunnel connection curve preferences {K25519MLKM768 CurveID(45074) CurveP256} connIndex=2 event=0 ip=198.41.200.23
May 31 02:04:44 ip-10-0-0-241.ec2.internal cloudflared[2636]: 2026-05-31T02:04:44 INF Registered tunnel connection connIndex=2 connection=Baf99fb-6029-4f84-a52-acac087454d7 event=0 ip=198.41.200.23 location=iad11 protocol=quic
May 31 02:04:44 ip-10-0-0-241.ec2.internal cloudflared[2636]: 2026-05-31T02:04:44 INF Tunnel connection curve preferences {K25519MLKM768 CurveID(45074) CurveP256} connIndex=3 event=0 ip=198.41.192.107
May 31 02:04:44 ip-10-0-0-241.ec2.internal cloudflared[2636]: 2026-05-31T02:04:44 INF Registered tunnel connection connIndex=3 connection=Bb8430e-A537-45f4-b64d-38d8b887502a event=0 ip=198.41.192.107 location=iad17 protocol=quic
lines 1-20/20 (END)
```

Error 1033

Ray ID: a042626ecbaa42dc • 2026-05-31 02:00:21 UTC

Cloudflare Tunnel error

What happened?

You've requested a page on a website (web.basitcloudlab.com) that is on the [Cloudflare](#) network. The host (web.basitcloudlab.com) is configured as a Cloudflare Tunnel, and Cloudflare is currently unable to resolve it.

What can I do?

If you are a visitor of this website:

Please try again in a few minutes.

If you are the owner of this website:

Ensure that cloudflared is running and can reach the network. You may wish to enable [load balancing](#) for your tunnel.

Was this page helpful?

Yes

No

← → ↻ 🔍 web.basitcloudlab.com

Basit Cloud Lab

Private AWS EC2 behind Cloudflare Tunnel

Built by Syed Basit Aftab

← → ↻ 🔍 web.basitcloudlab.com

Basit Cloud Lab

Private AWS EC2 behind Cloudflare Tunnel

Built by Syed Basit Aftab